

VoiceNoNG: Robust High-Quality Speech Editing Model without Hallucinations

Sung-Feng Huang^{1,2}, Heng-Cheng Kuo^{2,3}, Zhehuai Chen¹, Xuesong Yang¹, Pin-Jui Ku¹, Ante Jukic¹,
Chao-Han Huck Yang¹, Yu Tsao³, Yu-Chiang Frank Wang¹, Hung-yi Lee², Szu-Wei Fu¹

¹Nvidia, Taiwan

²Natioal Taiwan University, Taiwan

³Academia Sinica, Taiwan

{sungfengh, suzweif}@nvidia.com

Abstract

Voicebox and VoiceCraft are the current most representative models for non-autoregressive and autoregressive speech editing, respectively. Although both of them can generate high-quality speech edits, we identify their limitations: Voicebox is not good at editing speech with background audio, while VoiceCraft suffers from the hallucination-like problem. To maintain speech quality for varying audio scenarios and address the hallucination issue, we introduce VoiceNoNG, which combines the strengths of both model frameworks. VoiceNoNG utilizes a latent flow-matching framework to model the pre-quantization features of a neural codec. The vector quantizer in the neural codec provides additional robustness against minor prediction errors from the editing model, which enables VoiceNoNG to achieve state-of-the-art performance in both objective and subjective evaluations under diverse audio conditions. Audio examples and ablations can be found on the demo page¹ and supplementary material².

Index Terms: neural speech editing, flow-matching model, speech infilling

1. Introduction

In recent years, speech editing technology has rapidly gained prominence, offering substantial benefits across various domains. This technology enables users to seamlessly modify and enhance audio recordings [1], addressing issues such as slips of the tongue, mispronunciations, or transient background noise. Consequently, YouTubers and filmmakers can edit their speeches without re-recording, significantly boosting productivity and reducing costs. Below we first introduce recent speech editing techniques, and then summarize the contributions of this paper.

1.1. Speech editing techniques

In the early stages of speech editing, the process primarily involved cut-and-paste techniques to replace the speech segments that needed to be changed with the target speech segments. However, this approach often resulted in unnatural intonation and noticeable boundaries. Several works tried to address this issue with voice conversion and text-to-speech (TTS) techniques followed by manual manipulations with a user interface to ensure natural boundary transitions [2, 3]. Recently, as speech synthesis technology has become more advanced, methods for speech editing have generally been evolved into two main approaches: non-autoregressive speech infilling (inpainting) [4, 5, 6, 7, 8, 9], and autoregressive causal generation [10, 11, 12]. For the training of speech editing models, the models take masked audio along with its transcript as input, and the goal is to recover the

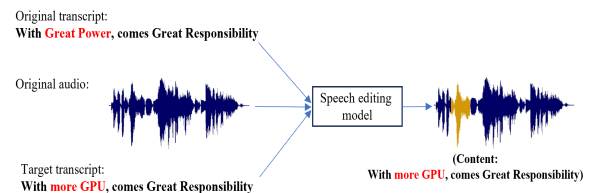


Figure 1: The input and output of speech editing models.

masked segments. During inference, the segment to be modified is simply masked, and the edited speech is generated according to the new target transcript. Similar to speech synthesis, the sequential dependency of autoregressive generation often results in better speech quality, while the parallel generation of non-autoregressive speech editing models allows for significantly faster inference. An overview of the input and output of current speech editing models are shown in Figure 1.

Among all these speech editing models, Voicebox [6] and VoiceCraft [12] are the two most representative non-autoregressive and autoregressive models respectively. Voicebox is a flow-matching-based generative speech model trained to perform speech infilling task by conditioning on both past and future audio contexts with text transcript, making Voicebox a versatile platform for various speech tasks, including zero-shot TTS synthesis, speech editing, transient noise removal, and style conversion. Furthermore, its flow-matching framework allows it to directly learn to transform noise distributions into audio data distributions, leading to more stable training, better generalization, and faster sample generation compared to traditional diffusion-based methods. Therefore, Voicebox outperforms previous state-of-the-art models while achieving up to 20 times faster performance in zero-shot TTS generation [6].

On the other hand, VoiceCraft [12] is an autoregressive model that fall under the Neural Codec Language Model (NCLM) category. NCLMs first convert audio into discrete tokens by a neural codec, which allows the model to learn better sequential token dependency through a next-token prediction task, and the predicted tokens can be further converted to high-quality audio using the neural codec decoder. For speech editing, the challenge lies in how to condition the causal generation on bidirectional context to infill the masked neural speech codec tokens according to the text transcript. VoiceCraft is therefore built upon a two-step token rearrangement procedure, comprising a causal masking step and a delayed stacking step, to enable autoregressive generation with bidirectional context with efficient multi-codebook modeling for high-quality generation.

¹<https://jasonswfu.github.io/NoNG-IS-/>

²<https://doi.org/10.5281/zenodo.15567613>

1.2. Contributions

Despite both Voicebox and VoiceCraft are successful in some speech editing conditions, we discover that they are not robust enough to generate promising speech editing under various scenarios. Therefore, in this paper, we first identify the main reasons for their limitations and propose a new SOTA speech editing method, VoiceNoNG, which combines their key strengths and eliminates those weaknesses. In summary, this paper makes several significant contributions:

- 1) We identify that the **hallucination-like** problem makes VoiceCraft difficult to follow the target transcription accurately. This leads to a higher word error rate (WER), indicating harder to follow the target transcription accurately.
- 2) We find that Voicebox **struggles to generate good edited speech with background audio**, and we identify the HiFi-GAN [13] vocoder might be the main reason for this problem.
- 3) We propose VoiceNoNG, a speech editing model that owns the key strengths of Voicebox and VoiceCraft, achieving **SOTA** performance in objective and subjective evaluations.
- 4) We find that the **vector quantizer** module in the neural codec is crucial for developing a robust speech editing model.

2. Proposed VoiceNoNG

A good speech editing model should precisely follow the target transcript while ensuring naturalness by maintaining the speaker’s characteristics and preserving the background audio (e.g., noise, music, etc.) in line with the surrounding context. In this section, we first discuss the drawbacks of Voicebox and VoiceCraft, and then explain how we address these issues.

2.1. Issues of Voicebox and VoiceCraft

In our preliminary experiment, we observe that Voicebox is **not good at generating speech edits with background audio**. We suppose there are two reasons. First, Voicebox is trained on LibriLight [14], which is a clean and large-scale audiobook corpus without background. Second, Voicebox utilizes a HiFi-GAN vocoder with Mel spectrograms as the audio feature, which has shown not generalizing well to various non-speech audio scenarios such as sound or music [15, 16]. A spectrogram example is shown in Figure 2, with audio provided on our demo website (section 2 of the website) and comprehensive experiments to verify this as presented in Section 3.

On the other hand, although VoiceCraft can obtain a better subjective preference, its WER is worse than the baseline (please refer to Table 4 of [12]). After carefully checking its generated speech, we identified that this higher WER is due to attention errors (hallucinations), which make it difficult for VoiceCraft to accurately follow the target transcription. Specifically, when the target transcript is complex (multiple instances of the same token), VoiceCraft **sometimes generates speech with missing or repeated words**. Recently, [17] also highlights that autoregressive-based TTS models have this kind of attention error, which often generates speech with repeating words, missing words, and misalignment. An example of attention errors is presented in Table 1, with audio samples provided on our demo website (section 3 of the website).

2.2. VoiceNoNG

To address the aforementioned issues, we propose VoiceNoNG, which aims to combine the strengths of both Voicebox and VoiceCraft. To avoid hallucination problems and enable the generation

Table 1: An example of VoiceCraft’s attention errors (hallucinations) in the RealEdit dataset. Words highlighted in red indicate missing or deleted words, while those in blue denote extra or inserted words. Parentheses signify simple substitution errors. Bolded texts indicate hallucinations (missing or repeated words).

Ground truth editing target:

“promise that you will not ask me to borrow **+any money from the bank for the** money of you for mister van brandt she rejoined and i **-will** accept your help gratefully”

transcript of VoiceCraft’s results :

“promise that you will not ask me to borrow any money **-from the bank for the money** of you **(-for +from)** mister van **(-brandt she +branch you)** rejoined and i accept your help gratefully”

Transcript of Proposed VoiceNoNG:

“promise that you will not ask me to borrow any money from the bank for the money of you for mister van **(-brandt +brant)** she rejoined and i **+will** accept your help gratefully”

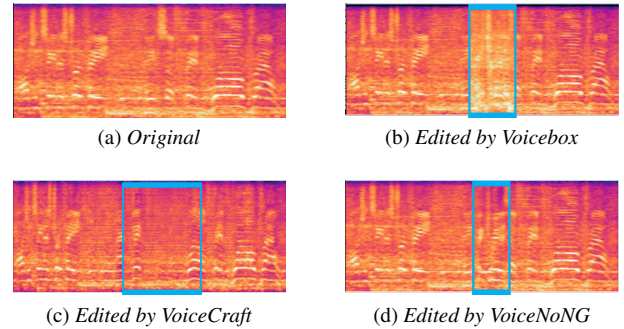


Figure 2: Issues of Voicebox and VoiceCraft. The original speech has noisy background, and the blue blocks indicate the generated segments. Voicebox generated loud and fuzzy sound, while VoiceCraft generated unwilling silence. Only VoiceNoNG produced clear speech with consistent background.

of speech with background audio, VoiceNoNG utilizes a latent flow-matching framework with neural audio codec.

2.2.1. Utilizing Descript Audio Codec (DAC)

Inspired by VoiceCraft, which trained their own EnCodec [18] as their speech tokenizer, we noticed that there are lots of recent works showing great success in generating speech and audio with neural audio codecs [16, 11, 12]. Therefore, we adopt the Descript Audio Codec (DAC) [19] as the acoustic feature for VoiceNoNG, which can be further converted back to audio with its paired audio codec decoder. More DAC details will be described in section 3.2.

Besides being trained on a diverse range of audio types and sources, another reason for using DAC is its ability to compress the features into a discrete latent space through a vector quantizer (VQ). This **provides additional robustness against minor prediction errors from the editing model** [20]. A small error, as long as it is not larger than the token decision boundary, will still be mapped to the correct token during VQ. This will be verified in Section 3.6.

2.2.2. Latent flow-matching model

Flow-matching-based non-autoregressive models [21, 16, 6, 22] and NCLM-based autoregressive models [23, 12, 11] are the two main model categories of high-quality speech generation for various tasks. However, since the hallucination-like problem exists for NCLM-based autoregressive speech models, we’re more willing to use the Voicebox-like flow-matching framework

for its controllability and faster inference speed.

In addition to the original flow-matching loss used in Voicebox, we apply a cross-entropy (CE) loss to further enhance our model’s codec prediction accuracy after VQ, which is inspired by NaturalSpeech 2 [24]. Since the codec token selection is based on the distance between the prediction and the candidates in the dictionary C , we can formulate this process as a probability distribution using the softmax operation. Consequently, the CE loss L_{ce} can be calculated as follows:

$$L_{ce} = -\frac{1}{|M|} \sum_{t \in M} \log\left(\frac{\exp(-\|(S(X)_t - Z_t)\|_2)}{\sum_{v=1}^V \exp(-\|(S(X)_t - C_v)\|_2)}\right) \quad (1)$$

where M represents the masked region, $S(\cdot)$ denotes our speech editing model with input features X , Z is the original codec token before masking, and V is the size of the dictionary. All vectors are L_2 -normalized before calculation. This loss function aims to minimize the distance between the prediction and the correct token and maximize the distance to the others.

3. Experiments

3.1. Training and testing sets for speech editing

Unless otherwise noted, we use GigaSpeech as the training corpus, as its diverse audio conditions enable our model to better perform background-preserving speech editing. For evaluating different speech editing models, we utilize the RealEdit dataset [12], a pioneering benchmark that features 310 manually-crafted examples reflecting real-world editing scenarios. This dataset includes tuples of original audio, original transcript, and edited transcript, meticulously crafted to ensure grammatical correctness and semantic coherence. RealEdit consists of 100 utterances from the LibriTTS dataset [25], 100 from YouTube (sourced from the GigaSpeech test set [26]), and 110 from the Spotify Podcast dataset [27]. The edits span various types—insertions, deletions, and substitutions—across different lengths, from short (1-2 words) to long (7-12 words), with single or multiple spans.

For each speech editing model, we generate edited speech for every tuple from the RealEdit dataset and then evaluate these generated speeches using the metrics introduced in the following subsections. During the objective evaluation, speech was generated with five different random seeds, and we reported the mean scores along with the standard deviation (shown in parentheses).

3.2. Speech editing model setup

Our proposed VoiceNoNG combines the strengths of both Voicebox and VoiceCraft, enabling our model to perform background-preserving speech edits without the hallucination issues. To verify that our modifications lead to the desired improvements, we compared our speech editing model with four other models: two VoiceCraft models (with sizes 330M and 880M), and two Voicebox models (using Mel-spectrograms as the acoustic feature, trained on LibriLight and GigaSpeech respectively). The results for VoiceCraft are based on the official code. Since no code and model checkpoints are available for Voicebox, we reproduced the results. Our Voicebox architecture settings are identical to those reported in the original paper, which features 330M parameters. For a better comparison, our VoiceNoNG shares the same architecture as Voicebox, except for the dimensions of the input and output acoustic features are changed according to DAC.

To delve into more details, DAC comprises three modules: an encoder that encodes the input waveform into a fixed-dimension vector sequence (pre-quantization feature), a residual

VQ module that discretizes the encoded representations into tokens, and a decoder that decodes the quantized representations (post-quantization feature) back into the waveform. In our experiments, we use the pre-quantization feature as our acoustic feature and apply an extra CE loss calculated against the ground-truth token labels to enhance the token prediction accuracy of VoiceNoNG. To demonstrate the impact of this setting, we also conduct an ablation study to highlight the benefits of modeling pre-quantization features and incorporating the CE loss.

3.3. Objective evaluation for speech intelligibility: WER

WER is an appropriate metric for evaluating TTS and speech editing since it provides a quantitative assessment of how accurately a system generates speech according to text transcript. Table 2 displays the WER results for various speech editing models using Whisper-large-v3 [28] as the ASR model. The WER is computed by comparing the transcribed text to the target transcript provided by the RealEdit datasets.

As indicated in the table, all flow-matching-based methods achieve lower WER than VoiceCraft, which is consistent with the findings presented in the VoiceCraft paper [12] and Table 1 that this is primarily due to the hallucination problem of VoiceCraft³. Additionally, the results demonstrate that when the training corpus for Voicebox is switched from LibriLight to GigaSpeech, the WER decreases across all types of audio sources, and further reducing the WER is achieved by changing the acoustic feature from Mel-spectrogram to DAC (our proposed VoiceNoNG).

We also perform an ablation study for VoiceNoNG to illustrate the advantages of modeling pre-quantization features and adding CE loss. When VoiceNoNG generates the pre-quantization features, it must be further processed through the VQ and DAC decoder to reconstruct the waveform, which offers additional robustness against prediction errors as noted in Section 2. In contrast, if VoiceNoNG outputs the post-quantization features (similar to [24]), only the DAC decoder is required for waveform reconstruction, which therefore yields the highest WER on average. Incorporating CE loss helps improve the token prediction accuracy, thereby also reduces the WER.

3.4. Objective evaluation for speech quality: SQUIM-(SI-SDR)

To assess the quality of the generated speech, we utilize a non-intrusive estimation of the scale-invariant signal-to-distortion ratio (SI-SDR), referred to as SQUIM-(SI-SDR) from [29]. The results of the estimated quality scores are presented in Table 3. It is evident that the SI-SDR for the original speech from LibriTTS is higher than that of the speech from YouTube and Spotify. This finding corresponds with the characteristics of the datasets, as the speech from YouTube and Spotify contains some background audio. Although the SI-SDR scores offer less clear information regarding the generated speech for YouTube and Spotify, we still provide the scores for reference.

In the case of LibriTTS, we can observe that the speech quality produced by VoiceNoNG is the best and closely resembles the original speech, while the quality score for the speech generated by Voicebox is the lowest, likely due to the limitations of the Mel-spectrogram and the vocoder used. We also observed that if VoiceNoNG utilized post-quantization features as the acoustic feature, the SI-SDR would significantly decrease. Furthermore, the standard deviations for Voicebox (Giga, Mel) and post-quantization are considerably larger, which aligns with our

³More examples can be found in our supplementary material and demo page.

Table 2: *WER evaluation of speech editing on the RealEdit dataset. An asterisk (*) indicates the reproduced model.*

	WER ↓ (%)			
	LibriTTS	YouTube	Spotify	Total
Original	1.84	6.13	5.87	4.65
VoiceCraft (330M)	3.72 (0.13)	7.41 (0.38)	4.63 (0.14)	5.23 (0.16)
VoiceCraft (830M)	3.77 (0.41)	7.36 (0.35)	5.43 (0.33)	5.52 (0.19)
Voicebox* (Libri, Mel)	3.64 (0.19)	6.26 (0.19)	5.45 (0.19)	5.13 (0.08)
Voicebox* (Giga, Mel)	3.48 (0.16)	6.03 (0.16)	5.36 (0.13)	4.97 (0.12)
VoiceNoNG	2.82 (0.20)	5.84 (0.21)	4.92 (0.21)	4.54 (0.14)
VoiceNoNG w/o CE loss	2.94 (0.23)	5.81 (0.18)	5.07 (0.13)	4.62 (0.09)
VoiceNoNG w/ post-quant.	3.16 (0.19)	5.59 (0.24)	5.38 (0.22)	4.73 (0.18)

Table 3: *SQUIM-(SI-SDR) evaluation of speech editing on the RealEdit dataset. We highlight the LibriTTS case in bold, as the scores for YouTube and Spotify provide less clear insights.*

	SI-SDR ↑			
	LibriTTS	YouTube	Spotify	Total
Original	23.94	19.55	19.22	20.85
VoiceCraft (330M)	22.22 (0.08)	19.97 (0.08)	20.15 (0.16)	20.76 (0.04)
VoiceCraft (830M)	22.53 (0.07)	20.04 (0.06)	20.26 (0.13)	20.92 (0.08)
Voicebox* (Libri, Mel)	19.49 (0.13)	17.96 (0.16)	16.32 (0.10)	17.88 (0.03)
Voicebox* (Giga, Mel)	18.10 (0.41)	16.86 (0.32)	15.83 (0.17)	16.90 (0.20)
VoiceNoNG	23.15 (0.09)	19.29 (0.05)	19.04 (0.08)	20.44 (0.05)
VoiceNoNG w/o CE loss	23.36 (0.09)	19.30 (0.09)	18.98 (0.06)	20.50 (0.04)
VoiceNoNG w/ Post-quant.	20.80 (0.26)	18.28 (0.11)	17.81 (0.10)	18.93 (0.05)

listening experience that these models are less robust and more likely to produce varying speech quality across the five random seeds. **This further validates the advantages of predicting pre-quantization features and employing the VQ module to correct minor prediction errors.**

3.5. Subjective evaluation

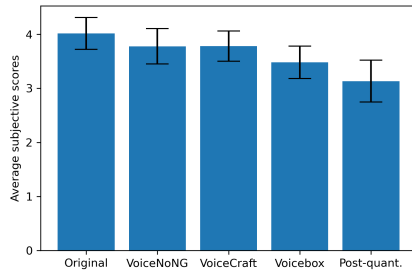


Figure 3: *Subjective scores of different speech editing methods.*

To subjectively evaluate the **audio quality** of various speech editing methods, we conducted a listening test using a 5-point scale. Participants listened to audio files, which could either be real or edited, and rated them on a scale where 5 represented highly natural and unaltered audio, while 1 indicated a strong belief that the audio had undergone partial editing (mainly follows [12]; please refer to Figure 5 in the supplementary material for detailed instructions). 8 original speeches were randomly selected from LibriTTS, YouTube, and Spotify, respectively. The corresponding edited speeches from VoiceCraft, Voicebox, the proposed VoiceNoNG, and VoiceNoNG with post-quantization were then chosen, resulting in a total of $(8 \times 3) \times 5 = 120$ utterances for each listener to evaluate. 15 listeners participated in the study. The experimental results are shown in Figure 3, revealing that the naturalness of speeches edited by VoiceNoNG and VoiceCraft is comparable to that of the original speech (we ask the listeners

to focus on audio quality while ignoring semantics, grammar, etc., therefore the hallucination issue of VoiceCraft may not be reflected). In contrast, the performance of VoiceNoNG with post-quantization exhibits a significant decline, warranting further analysis in the next section.

3.6. Robustness benefits from VQ module

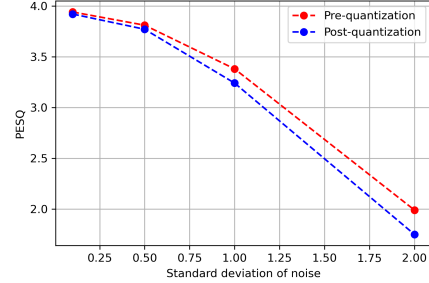


Figure 4: *Relationship between varying levels of noise added to pre-quantization and post-quantization DAC features and the PESQ scores of the resulting resynthesized speech.*

As shown in the previous experiments, the performance of VoiceNoNG declines when post-quantization features are used as learning targets. To validate the advantages of predicting pre-quantization features with VQ, we intentionally added Gaussian noise to these features to **simulate the prediction error incurred by the editing model**. LibriTTS subset from the RealEdit dataset was chosen for this experiment. In Figure 4, we present the PESQ scores [30] for resynthesized speech generated from the noisy DAC features at varying noise levels (the noise strength is first normalized based on the dynamic range of the input features). This figure illustrates the robustness benefits of VQ, showing an approximate improvement of 0.2 PESQ scores under moderate noise conditions.

4. Ethical Impact

The same capabilities that make speech editing valuable for legitimate purposes can be exploited to create sophisticated Deepfake audio [31], posing serious threats to public trust. Therefore, the research community is increasingly focused on building robust detection systems to identify manipulated speech [32]. Detailed experimental setup and results of anti-spoofing are described in our supplementary material, which demonstrates that the self-supervised pre-trained detector achieves near-perfect (i.e., 97%) utterance-level accuracy, even in more challenging conditions where non-edited segments are also resynthesized from the DAC.

5. Conclusion

In this study, we first examine the limitations of current advanced speech editing models. Voicebox produces lower-quality speech in the presence of background audio (e.g., noise, music), while VoiceCraft struggles to accurately follow text input, a common hallucination-like issue with autoregressive models. To address these challenges, we introduce VoiceNoNG, which leverages the strengths of both models and achieves SOTA performance in both objective and subjective evaluations (our demo page shows VoiceNoNG can keep the accent and even successfully generate background music for movie editing!). We also identify the vector quantizer module in the neural codec as crucial for achieving a robust speech editing model.

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